DATE	28-02-2026
TEAM ID	LTVIP2026TMIDS****
PROJECT NAME	HEMATOVISION: ADVANCED BLOOD CELL CLASSIFICATION USING TRANSFER LEARNING
MAXIMUM MARKS	4 MARKS

4.2 PROPOSED SOLUTION

The proposed solution for HematoVision: Advanced Blood Cell Classification Using Transfer Learning is to develop an automated and intelligent system that accurately classifies blood cell images using deep learning techniques. The system leverages transfer learning with pre-trained Convolutional Neural Network (CNN) models to improve performance while reducing training time and computational cost.

The solution begins with collecting a labeled dataset of microscopic blood cell images. These images undergo preprocessing steps such as resizing, normalization, and data augmentation to enhance image quality and improve model generalization. A pre-trained CNN model (such as ResNet, VGG, or MobileNet) is then fine-tuned on the blood cell dataset to learn domain-specific features effectively.

The trained model classifies different types of blood cells and provides predictions along with confidence scores. Performance is evaluated using metrics such as accuracy, precision, recall, and F1-score to ensure reliability and effectiveness.

Additionally, the system can be integrated with a user-friendly interface that allows laboratory technicians or medical professionals to upload blood cell images and instantly receive classification results. This makes the solution practical and suitable for real-world healthcare environments.

Overall, the proposed solution offers a fast, accurate, scalable, and cost-effective approach to blood cell classification, supporting early disease detection and improving the efficiency of medical diagnostics.